

6.4.5.3. Trusted Root CA Certificates

Each Node has one or more trusted Root CA certificates in its Node Operational credentials that it uses to verify ICA certificates and Node Operational Certificates presented by other Nodes, treating them as trust anchors as described in [RFC 5280](#). A Root CA certificate is self-signed. They are not verified but rather trusted because they were provisioned by a trusted Commissioner.

In the case where a Root CA issues the NOC, the Root CA certificate is used to attest to the validity of the NOC.

The trusted Root CA certificates that a Device trusts when the Device is verifying operational certificates are those stored in the [TrustedRootCertificates attribute](#) of that Device's [Operational Credentials cluster](#).

A device MAY have Root CA certificates that it trusts for purposes other than for operational credential verification. These certificates SHALL NOT appear in any Node's [TrustedRootCertificates attribute](#) of the [Operational Credentials cluster](#). The certificates configured in that cluster SHALL only be added during the commissioning process by the Commissioner, or during root rotation operations by an Administrator already trusted by the Node. Nodes SHALL NOT modify the [TrustedRootCertificates attribute](#) outside of the processing of [Operational Credentials cluster](#) commands.

The figures below show the Node Operational Certificate hierarchies, with and without optional ICAC.

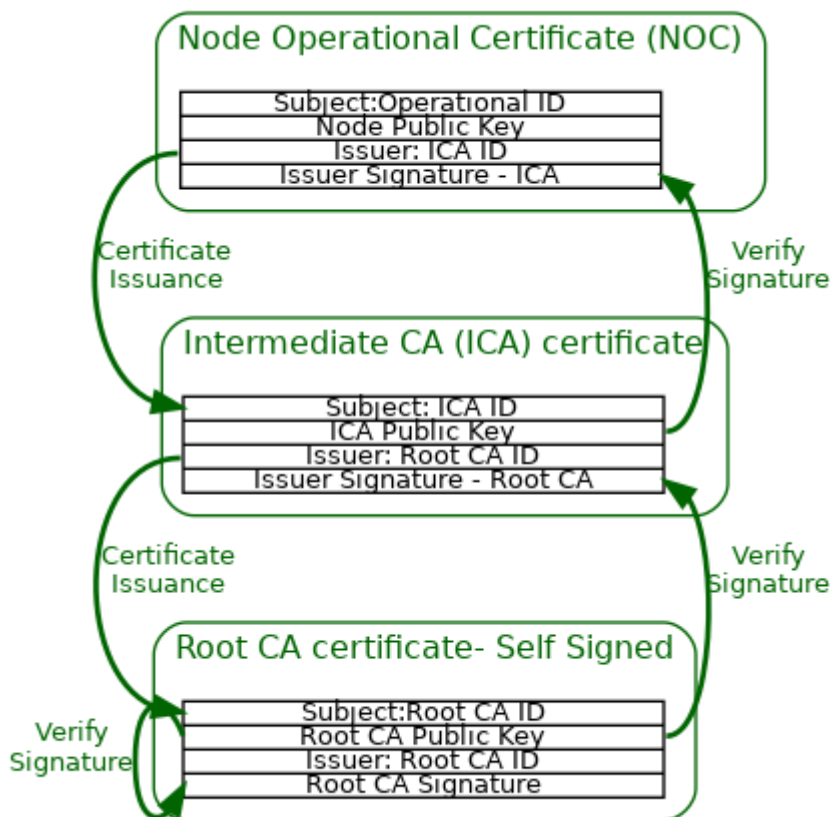


Figure 54. Node Operational Certificate PKI hierarchy with optional ICAC